

Digital Twin for Structural Monitoring and Anomaly Detection of the JRC Atmospheric Tower

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Abstract. Structural Health Monitoring is essential in detecting structural anomalies and enhancing the reliability of structures. Installation of sensors on existing and complex structures can be challenging, thus preventing the implementation of exhaustive monitoring systems. The development of structural digital twins, coupled with data from sensors, can contribute to more effective solutions, providing additional insights on the monitored structures. This work demonstrates the potential of digital twins for monitoring and identifying structural anomalies, by presenting the development of an application for a 100 *m*-tall steel truss atmospheric tower and base building. Operational Modal Analysis techniques, based on continuously monitored structural vibrations, are implemented to identify the dynamic parameters of the tower. This enables the possible detection of structural anomalies and changes in the towers modal response over time. The digital twin is then developed by combining real-time data with the virtual representation of the structure. We outline the details of the developed system, comprising the acquisition system, the digital platform, and the solution for the integration of the modular analysis components in the digital twin system architecture. The importance of digital twins in detecting structural anomalies during the life of the structure is here highlighted, demonstrating how it can enable proactive maintenance and contribute in reducing the risk of unexpected failures.

Keywords: Digital Twin · Dynamic Structural Response · Structural Health Monitoring.

1 Introduction

Modern society evolved through the building of complex and interconnected infrastructures that, while enabling greater efficiency and development, also

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present management and maintenance challenges to guarantee service reliability and prevent critical component failures. Concern over the structural integrity of high-rise structures, such as wind turbines, electricity transmission or telecommunications towers, has grown because of their widespread use and crucial role in networked infrastructures. These types of constructions are typically lattice steel structures, often exposed to severe environmental loads due to extreme winds and temperatures. Such loads, worsened by climate change, can lead to varied deflection shapes and potential failure locations. Additionally, long-term dynamic loads are particularly dangerous for such tall and slender structures inducing substantial dynamic behaviour that can lead to fatigue failure due to damage accumulation [10]. Lattice steel structures are constructed through the bolted assembly of steel members. A key concern in the maintenance of these structures is associated with bolt loosening issues, which can occur under various types of loading during their extended service life. As the bolts loosen, joint slips can occur, possibly leading to changes in the dynamic response of the tower and, ultimately, to an increased risk of failure [13][2, 7]. The adoption of Structural Health Monitoring (SHM) strategies based on sensor measurements in combination with Digital Twin (DT) solutions can provide valuable insights into the behaviour of these complex structures, enabling early detection of potential issues and informing proactive maintenance strategies to mitigate the risk of catastrophic failures. The numerical analysis in a DT accurate model could compensate for a small number of sensor installed on the physical counterpart [1]. This work aims to demonstrate how the concept of a DT can enable the collection, analysis, and aggregation of various sensor-collected data types in combination with numerical models of physical structures towards enhanced infrastructural maintenance strategies. This work presents an DT solution, based on modular analysis components and open-source tools, that integrates heterogeneous wireless sensors, a digital platform, and a modular Finite Element Method (FEM)-based architecture to enhance infrastructural maintenance strategies for a 100 *m*-tall atmospheric tower at the Joint Research Centre (JRC).

2 Digital Twins in Civil Infrastructures

The concept of DTs, initially developed in the digital transformation of Industry 4.0, is linked to up-to-date digital replica of actual physical assets [8, 11] to provide improved insights on the functioning of the real-world components towards enhanced operational efficiency, predictive maintenance strategies and simulations. A DT can combine real-time and historical sensor data from a physical asset, allowing the analysis of trends, patterns, and correlations that traditional monitoring methods may not be able to identify.

DTs applied to civil structures include SHM systems that are traditionally used for structural assessment and for the identification of issues through sensors monitoring and signal processing. However, the installation of sensors on existing complex structures like it is the case in existing bridges or tall structures can be challenging. When combined with numerical analysis of the digital twin

model, the ability to gain more information about the state of the structures from a small number of sensors can be a fundamental tool for preventing failures and providing guidance for maintenance interventions. Despite the promising benefits in DT developments and the increasing number of pilot applications and digital platforms, several challenges still prevent wide implementation and adoption. One of the primary obstacles to the development of DTs is the lack

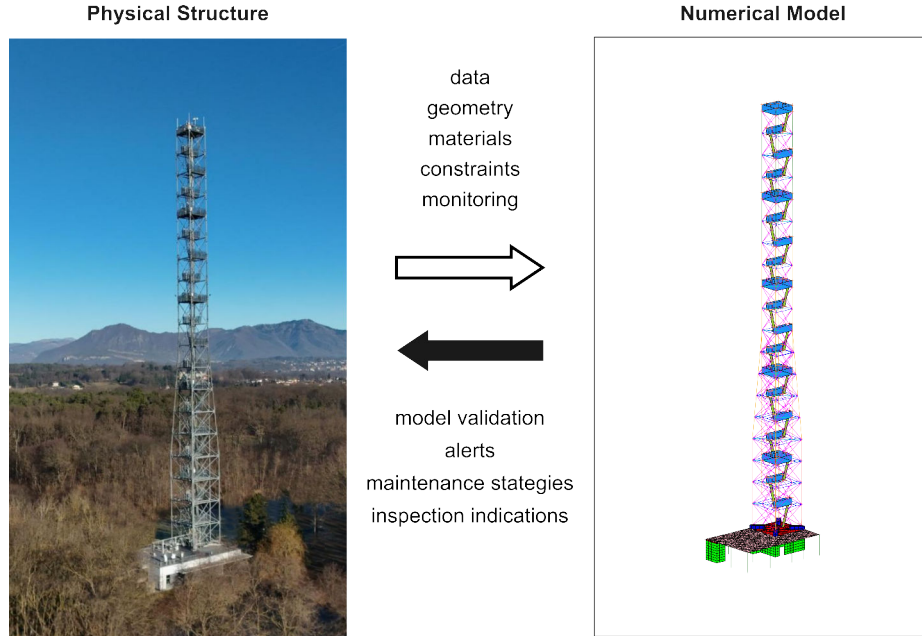


Fig. 1: The 100 *m*-tall JRC Atmospheric Tower and base building at the EC JRC in Ispra, Italy (left) and the correspondent FEM model adopted in the digital twin solution (right).

of data standards and interoperability for efficient information exchange across various platforms and systems. Interoperability is also crucial in the adoption of multiple analysis tools sharing common 3D models of the structure. Moreover, many infrastructures are equipped with legacy monitoring systems that require custom integration processes or that, in the case of proprietary solutions, may prevent access to collected data with different tools. Moreover, the creation of innovative heterogeneous sensing technologies for SHM requires innovative methods, to integrate data from multiple sources and the different technologies (e.g. satellite monitoring data), which enable sensing and data processing [11], such as cloud computing systems and Internet of Things (IoT) physical devices, in the development of DTs. Scalability and complexity present additional challenges in the implementation of DTs for infrastructures monitoring: the complex nature

of civil structures often leads to high computational demands and challenges in the scalability of such solutions, limiting the benefits of enhanced real-time monitoring and predictive analytics [9].

Another critical aspect is associated with the concept of a “true digital twin”, expressed in [12], that highlights the importance of achieving an accurate model and perform continuous updates to reflect the evolving condition of the physical structure over its lifetime to ensure a reliable DT representation. The integration of model updating techniques, with mathematical and procedural aspects presented in [5], is also envisaged in the developed DT architecture .

3 Case study

The JRC Atmospheric Tower (Fig. 1) is actually a 95 *m*-tall steel truss structure, supported by a 5 *m*-tall reinforced concrete building. The tower’s core consists of four vertical columns, arranged in a $4.75\text{ m} \times 4.75\text{ m}$ square, made of circular hollow sections with joints connected by 12 bolts. The core columns are surrounded by four outer columns, arranged in a $7.75\text{ m} \times 7.75\text{ m}$ square. At an elevation of 25 *m*, the outer columns are angled and converge to connect with the inner core. The floors are formed by a grid-work of HEA140 beams, with a floor every 5 *m*, and three alternating configurations.

Diagonal bracing tubes connect columns and floors. The stairs are made of inclined UPN 80 profiles, and the parapets consist of framed plates with a handrail and grilled plates.

The tower is anchored to the top of the base building through long bolts embedded in concrete shoes, with two sets of 12 anchorage bolts arranged in a circular pattern. The base building stands on a concrete foundation plate supported by 91 deep piles.

The elevator mast is a trussed column, formed by modular elements with a triangular plan, attached to the grid-work of beams at each floor level by a tube element and specific bolted plates. An elevator cabin, with a $1.2\text{ m} \times 1.2\text{ m}$ floor and a height of 2 *m*, can travel along the whole height of the tower with a combined weight of 650 *kg*.

4 Structural Monitoring System

The sensor monitoring system is designed to acquire vibration data from the tower continuously. The system consists of three solar-powered autonomous *G-link 200-OEM* wireless accelerometers (Fig. 2), with a detachable antenna and engineered enclosure, positioned at various tower levels (100 *m*, 80 *m*, 60 *m*). Additional sensors have recently been installed at the lower levels (40 *m* and 20 *m*). These sensors acquire accelerations along three orthogonal axes, with *z* aligned with the tower’s elevation, at a sampling rate of 64 *Hz* and employ a lossless communication mode that aims at preventing data packet loss due to interference. The sensor nodes use wireless communication with the network-connected *Microstrain WSDA-2000* gateway positioned at 60 *m*. In some cases,

the communication between the node and the base station might be difficult, especially when the Received Signal Strength Indication (RSSI) is low. A reliable communication can be reached for RSSI values between 0 and -75 dBm. A

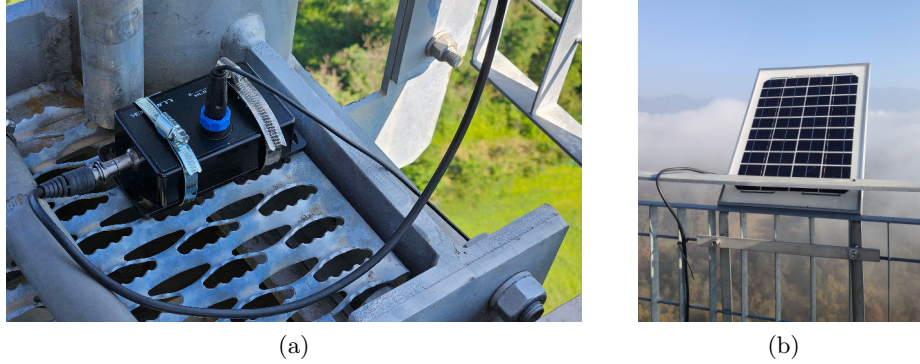


Fig. 2: Solar powered wireless accelerometer at 80 *m*: accelerometer sensor with detachable antenna (left) and solar panel (right).

custom programme, based on the MSCL Libraries, manages the gateway and the acquisition of all the data from the accelerometers mounted on the tower ensuring clock synchronisation across all the sensor nodes. Data packets are compiled and sent to an *IoT Agent* that forwards them to a centralised Context Broker [4]. The Context Broker is in charge of storing the data in the database; in the implemented solution, the daily acquisitions are stored in a primary database and then transferred to a timeseries database. This strategy ensures high performance of the system and efficient historical data visualisation. The SHM also integrates wind data measurements from sensors installed at 100 *m* and 40 *m* by the *JRC.C.5 Clean Air and Climate* unit, providing additional information of wind loads and directions.

5 Digital Twin System

The IT architecture (Fig. 3) developed at the JRC for the pilot application of the Atmospheric Tower DT system consists in *i*) the sensor monitoring and the data acquisition platform, *ii*) the SHM layer for signal processing and analysis, *iii*) the DT layer and *iv*) the visualisation tools. The whole DT platform is designed to integrate the sensor data with the FEM model, and to provide web applications for monitoring and analysing the tower's behaviour.

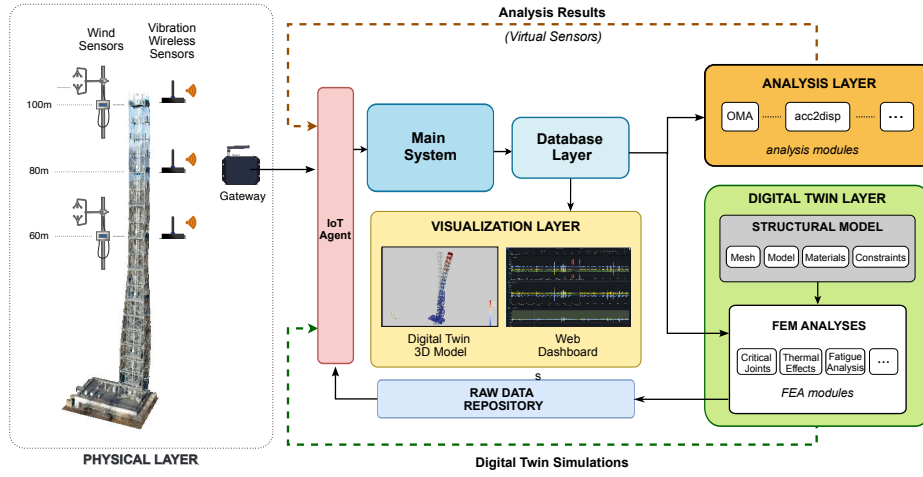


Fig. 3: Digital Twin system IT architecture.

5.1 Sensors Integration and Data Management

The physical sensors transmit wireless data streams to the Ethernet connected gateway. Received data are converted to Ultralight (UL) format, a generic lightweight text-based protocol, designed for devices and communications with limited resources (e.g. device memory or bandwidth). The UL messages are encapsulated in Message Queuing Telemetry Transport (MQTT) frames and forwarded to an *IoT Agent*, that serves as a bridge between the physical layer and the Context Broker (CB), the core hub of the platform which is used to integrate and manage the data and the measurements from the IoT sensors. We chose to use the open-source CB-based architecture for our application because it is widely used in European Smart City projects and is a component of the Connecting Europe Facility (CEF) building blocks. One of the goals of this work is also to assess such component for SHM applications and the integration with federated systems at larger scales. In addition, the CB is compliant with the NGSI-LD standard that allows to support the interoperability and the scalability within the same platform and across different smart city or digital systems.

The CB stores all the received daily data in a relational database. Older data are transferred to a timeseries database for improved performance in the management and visualisation of historical data. A web application allows the access and visualisation of all the sensor measurements, both recent and historical with custom dashboards.

5.2 Analysis Layer for Signal Processing

Signal processing is an essential part of SHM systems. In the implemented architecture, the analysis layer allows the implementation of multiple analysis

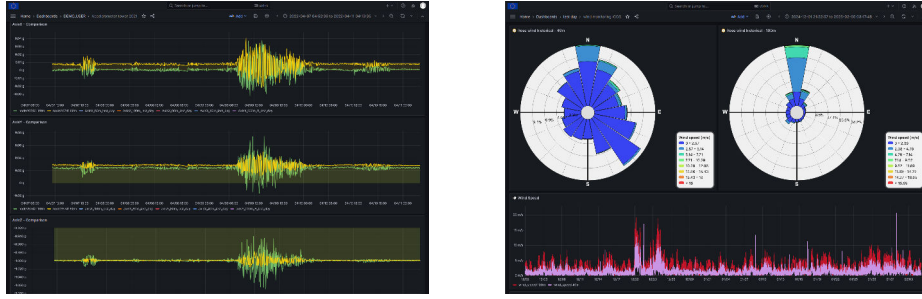


Fig. 4: Web dashboard with wind speed and vibrations (left) and wind sensors data (right).

scripts developed with multiple technologies. In this implementation, we integrate Python scripts to process vibration data acquired from the JRC Atmospheric Tower. Other algorithms (different scripting languages or compiled binary executables) can be easily implemented provided that a server-side engine is capable of executing the analysis software components. Two analysis components have been implemented at the moment for the analysis of the recordings of the installed accelerometers: *i*) a Fast Fourier Transform (FFT) vibration analyser suitable for real-time applications and diagnostics of the modal response of the structure over time, and *ii*) a vibration to displacement signal processing algorithm (Fig. 5). Acceleration data from the sensor nodes at the different levels

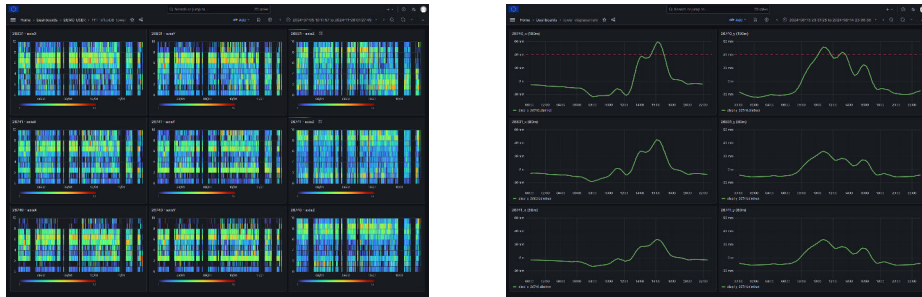


Fig. 5: Web dashboard with FFT variations (left) and daily computed displacement (right).

are filtered and processed to obtain estimates of the displacements. The results of the analyses are considered as *virtual sensors* of the system, and provide additional information of the structure through signal analysis. In adherence to the implemented IT architecture, virtual sensor data are sent to the *IoT Agent* for the storing in the database by the *Context Broker* and the visualisation in the web application dashboard.

5.3 Digital Twin Integration and Numerical Analyses

The digital twin application is developed to combine a Finite Element Analysis (FEA) component with real-world measurements from the installed sensors. In our implementation we rely on *Castem* FEM model [3]. For the pilot application we aim at applying real-world displacements, measured from the tower, to the FEM model in order to identify the most stressed joints. This pilot application analysis is representative of the processing and data flow that can be used to feed the FEM model with other inputs. The computed displacements are then made available to the FEM model for the analysis. *Castem* is executed cyclically in order to process the delivered data from the main system. At each execution, the latest data are loaded by the FEM program and applied to the model as imposed displacements. These are representative of the dynamics occurring on the real-world structure as a consequence mainly of wind loads and thermal deformations. The FEM analysis computes the stresses in the model's joints and exports all the transients as output. The output data from the FEM application is sent back to the main system via the *IoT Agent* to be ingested and stored in the main system databases, allowing quasi-immediate visualisation on the web application dashboard.

5.4 Finite Element Model Analysis Modules

The FEM model of the building is developed using *Castem* software and is fine-tuned to accurately reproduce the behaviour of the tower and its response under real loads and to perform simulations and in-depth analyses of structural components. The FEM model is crucial to compensate and complement for limited information from the installed sensors on the physical structure. The numerical model of the JRC Atmospheric Tower is composed of 5959 elements with 4884 nodes. The two parts of the structure (i.e., the base building and the steel truss tower) are modelled, although the displacements measured in correspondence of the concrete slab supporting the base of the tower are negligible. In order to have maximum flexibility in developing different analyses with the FEM model, the main information of the model, i.e. *i)* the mesh and *ii)* the materials, models and constraints, are stored separately. The DT layer allows for the storing of different FEA components; each module loads the general FEM structural model and performs specific analyses based on real-world measurements from sensors stored in the database layer. The main advantage of this approach is that multiple analyses can be performed independently with regular updates based on the measured dynamics of the real structure. Additional measurements from the numerical model are therefore possible and the results of simulations and analyses are also considered, as per the analysis layer performing signal analysis, as *virtual sensors* measurements. These are therefore transmitted to the *IoT Agent* for standardised processing in the system and storage in the database layer.

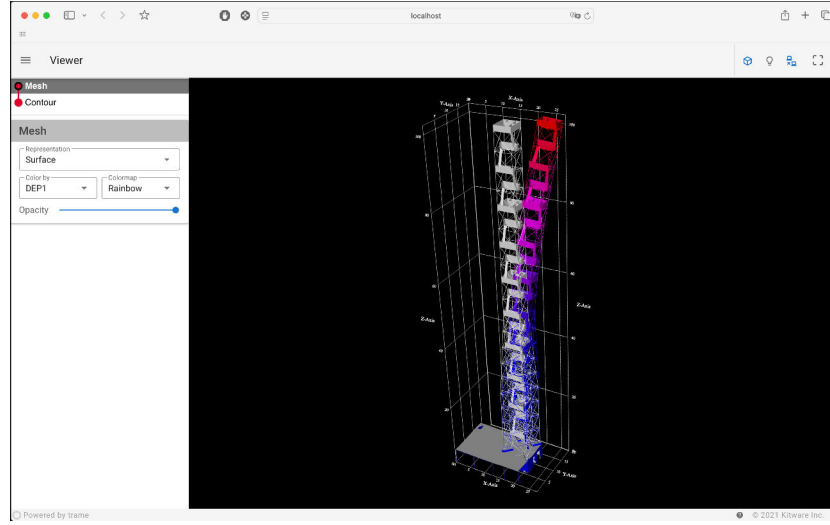


Fig. 6: Amplified deformed FEM JRC Tower 3D model visualisation based on virtual sensor displacements.

5.5 Historical Data and FEM 3D Visualisation

One fundamental aspect of the FEM analyses lays in the possibility to visualise the results in a 3D format with the option to display different variables and the deformed shape of the structure on the basis of the real-world measured structural response. An interactive 3D model is therefore essential for the post-processing evaluation by experts. In the designed solution, FEA results are saved from *Castem* in the widely used VTK format. This format is defined by *Visualisation Toolkit*, an open-source software software for processing and displaying scientific data, volume rendering, 3D graphics, and image processing.

The advantage of using open standards for the FEA is that results can be managed and visualised with multiple graphical applications. In the present DT architecture, a web application based on *Trame* [6], a Python framework for web-based interactive scientific visualisation of large data and 3D models offering possible integration with external Python libraries, VTK and *ParaView* backend, is integrated in the system. FEA output VTK files are saved by the multiple DT modules and then transferred to the open-source data management system (DMS). The DT 3D web application allows the listing and the selection of the desired output results and the online interactive visualisation or the raw data download for working with other compatible analysis and visualisation software.

An example application, developed for the JRC Atmospheric Tower DT, consists in the analysis of the deformations occurred over time. From the measured vibrations, the *acc2disp* module in the *analysis layer* calculates the displacements; the results are then sent back to the *IoT Agent* for the subsequent *Context Broker* processing and storing into the data repositories to be available

for and made available to FEM analyses. The FEM module *wind load deformations* loads the computed displacements at 100 m, 80 m and 60 m and applies the real-world measurements as imposed displacements. The results are saved as VTK file and the visualisation of the deformed shape with gradient colours correspondent to the displacement of each structural component can be displayed in the web application.

6 Conclusions

The development and implementation of a DT solution for the JRC Atmospheric Tower demonstrates the potential of this technology to provide flexible and scalable solutions in the field of SHM for complex infrastructures. By integrating real-time sensor data with numerical models and advanced analysis techniques, DT can provide in-depth insights into the behaviour of critical structures towards more effective strategies for maintenance, thus reducing the risk of unexpected failures. The proposed digital platform architecture demonstrates that DTs can be effectively implemented by integrating various open-source tools to monitor the structural response of tall structures, such as the JRC Atmospheric Tower. This approach enables the early detection of potential issues before they become critical through the integration of modular signal processing and FEM analysis components. Furthermore, the DT solution in this study addresses key challenges related to interoperability, data standardisation, and scalability in the development of DTs for infrastructure. Future research will focus on addressing challenges associated with the complete structured execution of all DT solution processes to develop a fully standardised framework and on the continuous validation and updating of DTs to ensure accurate and up-to-date numerical models, ultimately enhancing the safety, efficiency, and sustainability of critical infrastructure.

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